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Relay with arc extinguishing devices

Abstract

A relay includes a housing, a contact assembly and an auxiliary contact assembly. The contact assembly includes the movable contacts adapted to move towards or away from the stationary contacts under the driving of the movable contact bridge. The auxiliary contact assembly includes an auxiliary contact bridge and auxiliary contacts. At least portion of each of the auxiliary contacts extends into the housing. The auxiliary contact bridge moves synchronously with the movable contact bridge to be selectively and electrically connected to the auxiliary contacts. The housing includes a top surface spaced apart from the stepped portion in the movement direction of the movable contact bridge or the auxiliary contact bridge. The stationary contacts are arranged on the top surface of the housing. The portion of each of the auxiliary contacts arranged outside the housing is arranged on each of the stepped portions.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) The present disclosure is the National Stage of PCT International Application No. PCT/CN2021/090547, filed on Apr. 28, 2021, which claims priority to Chinese Patent Application No. “202021053107. X”, entitled “RELAY” and filed on Jun. 9, 2020 by BYD Company Limited. The entire contents of the above-referenced applications are incorporated herein by reference.

FIELD

(2) The present disclosure relates to the technical field of relays, and more specifically to a relay.

BACKGROUND

(3) In the related art, for a relay with auxiliary contacts, the on-off states of high-voltage main contacts are monitored mainly by applying low-voltage sampling signals to the auxiliary contacts through the on-off states of the low-voltage auxiliary contacts so as to learn whether the main contacts are normally switched on and off, and whether a fault phenomenon such as closed non-conduction or contact sintering occurs.

(4) At present, in the market, the common relay with reed structure has a simple structure, but the distance between the auxiliary contact and the main contact is too close, which increases the risk of high and low voltage breakdown between the main contact and the auxiliary contact when the contact is arcing with load.

SUMMARY

(5) The present disclosure is to resolve at least one of the technical problems in the related art. To this end, one objective of the present disclosure is to provide a relay with relatively high safety in use.

(6) The relay according to an embodiment of the present disclosure includes a housing, a contact assembly, and an auxiliary contact assembly. The contact assembly includes movable contacts, stationary contacts and a movable contact bridge. The movable contacts are adapted to move towards or away from the stationary contacts under the driving of the movable contact bridge. The auxiliary contact assembly includes an auxiliary contact bridge and auxiliary contacts. At least portion of each of the auxiliary contacts extends into the housing. The auxiliary contact bridge moves synchronously with the movable contact bridge to be selectively and electrically connected to the auxiliary contacts. The housing includes a top surface and stepped portions. The top surface is spaced apart from the stepped portion in the movement direction of the movable contact bridge or the auxiliary contact bridge. The stationary contacts are arranged on the top surface of the housing. A portion of each of the auxiliary contacts arranged outside the housing is arranged on each of the stepped portions.

(7) According to the relay of the embodiment of the present disclosure, by providing the auxiliary contacts and making the portions of the auxiliary contacts outside the housing be spaced apart from the stationary contacts, on the premise of making the detection of the operating state of the contact assembly safer and more reliable, the occurrence of high and low voltage breakdown between the

auxiliary contacts and the stationary contacts can be avoided, and the working stability of the relay can be improved.

(8) According to some embodiments of the present disclosure, an area of the housing between two adjacent stationary contacts is formed as a stepped portion, and the auxiliary contacts corresponding to the two adjacent stationary contacts are all arranged in the stepped portion.

(9) Further, the stepped portion is formed on a side wall of the housing, and the two auxiliary contacts corresponding to the two adjacent stationary contacts are arranged on the same stepped portion.

(10) In some embodiments, the number of the stationary contacts is multiple, the multiple movable contacts electrically connected to each other are arranged on the movable contact bridge and correspond to the multiple stationary contacts on a one-to-one basis, and the movable contact bridge can selectively move towards the stationary contacts so as to conduct at least two of the multiple stationary contacts; and the number of auxiliary contacts is multiple, and the multiple auxiliary contacts correspond to the multiple stationary contacts on a one-to-one basis, and each of auxiliary contacts extends out of the housing and is far away from the stationary contact

(11) According to some embodiments of the present disclosure, the relay further includes arc extinguishing devices. The number of the arc extinguishing devices is multiple. Each of the arc extinguishing devices is arranged on an outer side of the corresponding stationary contact and includes a cover and a magnetic member. The magnetic member is arranged on at least one of two side walls of the cover facing each other.

(12) Further, the relay includes a driving member. The driving member is connected to the movable contact bridge and is adapted to move the movable contacts towards or away from the stationary contacts.

(13) Optionally, a first armature is arranged on the movable contact bridge, a second armature is arranged in the accommodating groove of the housing, and the first armature and the second armature are adapted to be magnetically attracted when the movable contact conducts the corresponding stationary contact.

(14) In some embodiments, the driving member includes a fixed seat, an elastic member and a driving rod, one end of the elastic member is fixed on the fixed seat, the other end of the elastic member pushes against the movable contact, and the driving rod is fixed on the fixed seat to push the fixed seat to move.

(15) Further, the relay includes a fixed support, the elastic member is arranged in the fixed support, a lower end of the fixed support is connected to the fixed seat, a guide groove is formed in an upper end of the fixed support, and a guide block cooperating with the guide groove is arranged on the first armature.

(16) Further, the fixed seat is configured as an insulator.

(17) Additional aspects and advantages of the present disclosure are partially provided in the following description, and partially become apparent in the following description or understood through the practice of the present disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The above and/or additional aspects and advantages of the present disclosure will become evident and easy to understand through embodiments described with reference with drawings outlined below:

(2) FIG. 1 is a schematic cross-sectional view of a relay according to an embodiment of the present disclosure;

(3) FIG. 2 is another schematic view of a relay according to an embodiment of the present

disclosure;

(4) FIG. 3 is a schematic view of a housing of a relay according to an embodiment of the present disclosure;

(5) FIG. 4 is a cooperation schematic view of an arc extinguishing device and a housing of a relay according to an embodiment of the present disclosure;

(6) FIG. 5 is a cooperation schematic view of movable contacts and a driving member of a relay according to an embodiment of the present disclosure; and

(7) FIG. 6 is a schematic top view of an arc extinguishing device of a relay according to an embodiment of the present disclosure.

REFERENCE NUMERALS OF THE ACCOMPANYING DRAWING

(8) Relay **100**, Relay body **10**, stationary contact **11**, movable contact **12**, housing **13**, accommodating groove **131**, stepped portion **132**, driving member **14**, fixed seat **141**, elastic member **142**, driving rod **143**, fixed support **15**, guide groove **151**, auxiliary contact **16**, movable contact bridge **17**, first armature **18**, guide block **181**, arc extinguishing device **20**, cover **21**, first end portion **211**, second end portion **212**, connecting portion **213**, and magnetic member **22**.

DETAILED DESCRIPTION

(9) Embodiments of the present disclosure are described in detail below, and examples of the embodiments are shown in accompanying drawings, where the same or similar elements or the elements having same or similar functions are denoted by the same or similar reference numerals throughout the description. The embodiments that are described with reference to the accompanying drawings are exemplary, and are only used to interpret the present disclosure, instead limiting the present disclosure.

(10) A relay **100** according to an embodiment of the present disclosure is described below with reference to FIGS. 1-6.

(11) As shown in FIGS. 1, 2 and 3, the relay **100** according to an embodiment of the present disclosure includes: a housing **13**, a contact assembly and an auxiliary contact assembly.

(12) Among others, the contact assembly includes: movable contacts **12**, stationary contacts **11** and a movable contact bridge **17**, and the movable contacts **12** are adapted to move towards or away from the stationary contacts **11** under the driving of the movable contact bridge **17**.

(13) The auxiliary contact assembly includes: an auxiliary contact bridge (not shown in the figures) and auxiliary contacts **16**, at least portion of each of the auxiliary contacts **16** extends into the housing **13**, and the auxiliary contact bridge moves synchronously with the movable contact bridge **17** to be selectively and electrically connected to the auxiliary contacts **16**.

(14) In particular, the auxiliary contact bridge is adapted to move synchronously with the movable contact bridge **17** under the driving of the driving member **14**. Thus, when the movable contacts **12** are electrically connected to the stationary contacts **11**, the auxiliary contact bridge moves synchronously towards the auxiliary contacts **16** to conduct the auxiliary contact bridge and the auxiliary contacts **16**.

(15) It will be appreciated that the housing **13** includes: a top face and stepped portions **132**, the top surface is spaced apart from the stepped portions **132** in the movement direction of the movable contact bridge **17** or the auxiliary contact bridge, the stationary contacts **11** are arranged on the top surface of the housing **13**, and a portion of each of the auxiliary contacts **16** arranged outside the housing **13** is arranged on each of the stepped portions **132**.

(16) Specifically, when the contact assembly is in a conduction position (high-voltage conduction), the auxiliary contact bridge is electrically connected to the auxiliary contacts **16**, so that the auxiliary contacts **16** are also conducted (low-voltage conduction), the contact assembly is broken, and the auxiliary contact assembly is also broken, that is, the communication state between the contact assemblies can be determined by detecting the connection state between the auxiliary contacts **16**.

(17) When the contact assembly is worn and cannot be normally conducted (at a conduction

position, but the movable contact bridge **17** cannot be in contact with the stationary contacts **11**) and cannot be normally disconnected (the movable contact bridge **17** and the stationary contacts **11** are sintered or arcing is formed between the movable contacts **12** and the stationary contacts **11**), the state of the contact assembly can be determined according to the conduction condition of the auxiliary contacts **16** and the conduction condition of a circuit where the auxiliary contact assembly is.

(18) Furthermore, the auxiliary contacts **16** are arranged on the stepped portions **132**, and the stationary contacts **11** are arranged on the top surface of the housing **13**, so that the auxiliary contacts **16** and the stationary contacts **11** are spaced apart in the movement direction of the movable contact bridge **17** or the auxiliary contact bridge, so as to ensure that the distance between the auxiliary contacts **16** and the stationary contacts **11** always meets the use requirements during the operation of the relay **100**, so as to avoid a continuous arc between the auxiliary contacts **16** and the stationary contacts **11** and avoid a high-low voltage breakdown phenomenon.

(19) According to the relay **100** of the embodiment of the present disclosure, by arranging the auxiliary contacts **16** and making the portions of the auxiliary contacts **16** being outside the housing **13** be spaced apart from the stationary contacts **11**, on the premise of making the detection of the operating state of the contact assembly safer and more reliable, the occurrence of high and low voltage breakdown between the auxiliary contacts **16** and the stationary contacts **11** can be avoided, and the working stability of the relay **100** can be improved.

(20) According to the relay **100** of the embodiment of the present disclosure, the number of the stationary contacts **11** is multiple, and the multiple stationary contacts **11** are arranged on the housing **13**; the movable contact bridge **17** is provided with multiple movable contacts **12** electrically connected to each other, the multiple movable contacts **12** correspond to the multiple stationary contacts **11** on a one-to-one basis, and the movable contact bridge **17** can selectively move towards or away from the stationary contacts **11** so as to conduct at least two of the multiple stationary contacts **11**; and the number of auxiliary contacts **16** is multiple, and the multiple auxiliary contacts **16** correspond to the multiple stationary contacts **11** on a one-to-one basis, and each of the auxiliary contacts **16** extends out of the housing.

(21) Specifically, the movable contact bridge **17** is adapted to drive the movable contacts **12** to move towards the stationary contacts **11** so as to conduct at least two of the multiple stationary contacts **11** via at least two of the multiple movable contacts **12** electrically connected to each other (namely, high-voltage circuits where the movable contacts **12** and the stationary contacts **11** are arranged are conducted); the auxiliary contacts **16** and the stationary contacts **11** are correspondingly arranged, the auxiliary contacts **16** extend out of the housing **13**, and when the high-voltage circuits where the stationary contacts **11** and the movable contacts **12** are arranged are conducted, the auxiliary contacts **16** corresponding to the stationary contacts **11** are connected to detection circuits (the detection circuits are low-voltage circuits), so that the detection of the high-voltage circuits where the movable contacts **12** and the stationary contacts **11** are arranged is achieved by detecting interfaces of the auxiliary contacts **16** in the low-voltage circuits.

(22) According to the relay **100** of the embodiment of the present disclosure, the detection of the operating state of the high-voltage circuits where the stationary contacts **11** and the movable contacts **12** are arranged is made safer and more reliable by arranging the auxiliary contacts **16** and arranging the auxiliary contacts **16** outside the housing **13**, furthermore, the connection of the auxiliary contacts **13** and detection components is made simpler and more convenient by arranging the auxiliary contacts **16** outside the housing **13**, and thus, the detection of the operating state of the relay **100** is made simpler and more convenient.

(23) It is to be understood that the stationary contacts **11** may be configured as the multiple stationary contacts sequentially connected in parallel or the multiple stationary contacts connected in series.

(24) As shown in FIG. 4, an area of the housing **13** arranged between two adjacent stationary

contacts **11** is formed as a stepped portion **132**, and the auxiliary contacts **16** corresponding to the two adjacent stationary contacts **11** are all arranged in the stepped portion **132**, and the stepped portion **132** is spaced apart from the top surface of the housing **13** and is adjacent to the bottom surface of the housing **13**.

(25) In other words, the stepped portion **132** is formed on the side wall of the housing **13**, the two auxiliary contacts **16** corresponding to the two adjacent stationary contacts **11** are arranged on the same stepped portion **132**, the upper end of each stationary contact **11** protrudes from the upper end surface of the corresponding cover **21**, each of the auxiliary contacts **16** is arranged the corresponding stepped portion **132**, and the stepped portion **132** is spaced apart from the top surface of the housing **13** in the height direction.

(26) In this way, on the one hand, the upper end of each stationary contact **11** is made to protrude from the upper end surface of the cover **21** so as to facilitate the electrical connection of surrounding components electrically connected to the stationary contact **11** and the stationary contact **11**, and in one embodiment, the working stability of the relay **100** is improved, while the height of the relay **100** can also be reduced; on the other hand, a continuous arc between the auxiliary contact **16** and the stationary contact **11** can be avoided, the high-low voltage breakdown is avoided, and the working stability and use safety of the relay **100** are improved. As shown in FIG. **1** and FIG. **2**, the number of arc extinguishing devices **20** is multiple, each arc extinguishing device **20** is arranged on an outer side of the corresponding stationary contact **11** and includes a cover **21** and a magnetic member **22**, and the magnetic member **22** is arranged on at least one of two side walls of the cover **21** facing each other.

(27) That is to say, one arc extinguishing device **20** is arranged outside each stationary contact **11**, and an electric arc generated when the movable contact **12** is separated from the stationary contact **11** can be extinguished by the corresponding arc extinguishing device **20**. It should be noted that electric arc is a gas discharge phenomenon, which has a certain hazard to electric appliances. The arc extinguishing means that the electric arc is elongated under the action of the magnetic field force of the magnetic field of the arc extinguishing device **20** to extinguish the electric arc for solving the influence of the electric arc in the electric appliance. If the electric arc is a direct current electric arc, after the electric arc is elongated, the static volt-ampere characteristic moves up, and the electric arc can be extinguished; and if the electric arc is an alternating current electric arc, after the electric arc is elongated, reignition of the electric arc is difficult due to an increase in arcing voltage.

(28) The magnetic member **22** is arranged on at least one of the two side walls of the cover **21** facing each other and is between the side walls of the cover **21** facing each other, which can make the magnetic field be generated in the cover **21**, and the electric arc generated when the movable contact **12** is separated from the stationary contact **11** is pulled away from the movable contact **12** and the stationary contact **11** under the action of the magnetic field force, so as to elongate the electric arc to accelerate arc extinction and improve the arc extinction effect of the relay **100**.

(29) In one embodiment, the cover **21** includes a first end portion **211**, a second end portion **212**, and a connecting portion **213** connecting the first end portion **211** and the second end portion **212**, a side wall of the first end portion **211** is directly opposite to a side wall of the second end portion **212**, the magnetic member **22** is arranged on the side wall of the first end portion **211**, the magnetic member **22** is arranged between the first end portion **211** and the second end portion **212**, and one side of the magnetic member **22** facing away from the side wall of the second end portion **212** is in contact with the first end portion **211**, so that the second end portion **212** forms a polarity different from a polarity of the magnetic member **22** facing the second end portion **212**.

(30) For example, one side of the magnetic member **22** facing the second end portion **212** is N-pole, one side of the magnetic member **22** contacting with the first end portion **211** is S-pole, the second end portion **212** forms the polarity of the S-pole, a magnetic field in which magnetic induction lines face the second end portion **212** from the magnetic member **22** is formed in the

cover **21**, the stationary contact **11** is in the magnetic field, and when the movable contact **12** is separated from the stationary contact **11**, the generated electric arc is elongated under the action of the magnetic field force, and the arc extinguishing is accelerated.

(31) It will be appreciated that the arrangement of only one magnetic member **22** in the cover **21** is an embodiment of the present disclosure and that the arrangement of the magnetic members **22** at both ends of the cover **21** can increase the strength of the magnetic field.

(32) In the embodiment shown in FIG. **6**, the two magnetic members **22** are respectively arranged on the side walls of the first end portion **211** and the second end portion **212**, which are opposite to each other, and the opposite sides of the two magnetic members **22** have different polarities, thereby increasing the magnetic field strength in the cover **21** compared to the case where only one magnetic member **22** is arranged. In addition, the cover **21** has a magnetism gathering effect, and the cover **21** in combination with the magnetic member **22** further increases the magnetic field intensity in the cover **21** to improve the arc extinguishing effect of the arc extinguishing device **20**.

(33) The directions of the electric currents of the two electrically connected stationary contacts **11** are respectively inflow and outflow, and the directions of the magnetic induction lines formed between the two magnetic members **22** in the two arc extinguishing devices **20** are the same, so that the directions of the magnetic field forces acting on the electric arcs generated in the two arc extinguishing devices **20** are away from each other, the electric arcs in the two arc extinguishing devices **20** move in opposite directions, the electric arcs are elongated and are rapidly extinguished, meanwhile, the relay **100** has no polarity, and the use safety and adaptability are improved.

(34) As shown in FIG. **6**, according to some embodiments of the present disclosure, the directions of the magnetic induction lines formed between the two magnetic members **22** in the multiple arc extinguishing devices **20** are the same. Thus, the directions of electric currents of the two stationary contacts **11** connected via the movable contact **12** are opposite, and when electric currents are generated in the two stationary contacts **11**, the magnetic fields in which the two stationary contacts **11** are located respectively generate magnetic flux forces with equal magnitude and opposite directions, so that the electric arcs generated on the two stationary contacts **11** respectively move in opposite directions under the action of the magnetic flux forces, so as to effectively avoid the occurrence of a continuous arcing in the electric arcs, make the cut-off effect of the relay **100** better, and improve the use safety of the relay **100**.

(35) As shown in FIG. **3**, in some embodiments, the relay body **10** further includes the housing **13**, at least portion of the housing **13** is arranged on the periphery of the multiple stationary contacts **11**, the housing **13** is provided with an accommodating groove **131** between the two adjacent stationary contacts **11**, and at least portion of the cover **21** extends into the accommodating groove **131** so as to be fixed on the housing **13**.

(36) In other words, the housing **13** is provided with multiple through-holes, one stationary contact **11** is arranged in each through-hole, and the stationary contact **11** is in sealed connection with the housing **13**; furthermore, the accommodating groove **131** is arranged between two adjacent stationary contacts **11**, so that the cover **21** of the arc extinguishing device **20** corresponding to each stationary contact **11** can protrude into the accommodating groove **131**, thereby improving the fixing effect of the arc extinguishing device **20** on the housing **13**, and making the structure of the relay **100** more reasonable and the arc extinguishing effect of the multiple arc extinguishing devices **20** fixed on the housing **13** more stable and reliable.

(37) In addition, it should be noted that multiple gas flow chambers communicating with each other are correspondingly formed inside the housing **13** by arranging the accommodating groove **131**, gas can flow inside the multiple gas flow chambers, however, metal substances that are electrically ablated are isolated outside the housing **13**, the insulation property and the voltage withstand performance of the relay **100** after a damage test of the metal substances caused by an electrical life, a short circuit, etc. are improved, and at the same time, the distance between the two adjacent stationary contacts **11** is made further by the accommodating groove **131**, the creepage distance is

increased, and the arc extinguishing effect can be improved in one embodiment.

(38) In some embodiments, the multiple stationary contacts **11** include: positive stationary contacts and negative stationary contacts, the cover **21** outside the positive stationary contacts and the cover **21** outside the negative stationary contacts are attached to each other, and the attached side walls of the two covers are arranged in the accommodating groove **131**.

(39) In other words, among the multiple stationary contacts **11**, the covers **21** of the adjacent positive stationary contact and negative stationary contact are arranged together in one accommodating groove **131**, so that, on the one hand, the number of grooves in the housing **13** can be reduced, the production cost of the housing **13** can be lower, and the structural strength of the housing **13** can be higher, and on the other hand, the two adjacent covers **21** can be attached to the same accommodating groove **131** so as to improve the fixing effect of the adjacent arc extinguishing devices **20** in the accommodating groove **131**.

(40) As shown in FIG. 5, the relay body **10** further includes a driving member **14**, the movable contacts **12** and the stationary contacts **11** are arranged oppositely and are spaced apart, and the driving member **14** is connected to the movable contacts **12** and is adapted to move the movable contacts **12** towards or away from the stationary contacts **11**.

(41) Specifically, a sealed cavity is defined inside the housing **13**, the movable contacts **12** and at least portion of the driving member **14** are arranged in the sealed cavity, the movable contacts **12** are spaced apart from the stationary contacts **11** in an initial state to maintain a certain electric clearance, and the movable contacts **12** can move towards or away from the stationary contacts **11** under the driving of the driving member **14**, so that the on-off control of the relay **100** is simpler and more convenient.

(42) In one embodiment, the relay **100** includes first armatures **18** and second armatures **23**, each of the first armatures **18** is arranged on one side of each of the movable contacts **12** facing away from each of the stationary contacts **11**, each of the second armatures **23** is arranged on one side of each of the movable contacts **12** facing each of the stationary contacts **11**, and the first armatures **18** and the second armatures **23** are adapted to generate a magnetic attraction force when the movable contacts **12** conduct the corresponding stationary contacts **11** so as to press the movable contacts **12** against the corresponding stationary contacts **11**.

(43) In other words, each of the first armatures **18** is arranged on one side of each of the movable contacts **12** facing away from each of the stationary contacts **11**, each of the second armatures is arranged on one side of each of the movable contacts **12** facing each of the stationary contacts **11**, and the first armatures **18** and the second armatures are adapted to be magnetically attracted when the movable contacts **12** conduct the corresponding stationary contacts **11**, and push against the movable contacts **12** by the magnetic attraction force so as to resist the repulsive force between the movable contacts **12** and the stationary contacts **11**. In this way, in one embodiment, the working stability of the relay **10** can be improved, and the conduction effect between the movable contacts **12** and the stationary contacts **11** can be better.

(44) In some embodiments, each of the first armatures **18** is arranged on one side of each of the movable contacts **12** facing away from each of the stationary contacts **11**, the second armatures are arranged in the accommodating grooves **131** of the housing **13**, and the first armatures **18** and the second armatures are adapted to be magnetically attracted when the movable contacts **12** conduct the corresponding stationary contacts **11**.

(45) Specifically, at least portion of the movable contact **12** is arranged between the first armature **18** and the second armature; when the movable contact **12** conducts the corresponding stationary contact **11**, the first armature **18** and the second armature are magnetized, and the sides of the first armature **18** and the second armature facing each other are magnetically opposite, and generate a magnetic attraction force, which is formed as a force to push against the movable contact **12** to move towards the stationary contact **11**, thereby improving the contact pressure between the stationary contact **11** and the movable contact **12**, making the contact stability between the movable

contact **12** and the stationary contact **11** higher, and improving the working operational stability of the relay **100**.

(46) It should be understood that “the arrangement of the first armature **18** on one side of the movable contact **12** away from the stationary contact **11**, and the arrangement of the second armature in the accommodating groove **131**” is merely one embodiment of the present disclosure, and in other embodiments, the second armature may not be arranged in the accommodating groove **131**.

(47) For example, the driving member **14** includes a driving device and a driving rod **143**; the driving device drives the movable contact **12** to move towards or away from the stationary contact **11** by driving the driving rod **143**; the driving rod **143** passes through the first armature **18**, a movable contact plate **12** and the second armature in sequence, so that the first armature **18** and the second armature are respectively arranged at two sides of the movable contact **12**; and when the movable contact **12** conducts the corresponding stationary contact **11**, the first armature **18** and the second armature are magnetized, and the sides of the first armature **18** and the second armature facing each other are magnetically opposite, and generate a magnetic attraction force, which is formed as a force to push against the movable contact **12** to move towards the stationary contact **11**, thereby improving the contact pressure between the stationary contact **11** and the movable contact **12**.

(48) The present disclosure does not define the specific arrangement positions of the first armature **18** and the second armature, and as long as the first armature **18** and the second armature are respectively at two sides of the movable contact **12**, when the movable contact **12** conducts the corresponding stationary contact **11**, the contact pressure between the stationary contact **11** and the movable contact **12** can be improved.

(49) In the particular embodiment shown in FIG. 5, the driving member **14** includes a fixed seat **141**, an elastic member **142** and a driving rod **143**, one end of the elastic member **142** is fixed on the fixed seat **141**, the other end of the elastic member **142** pushes against the movable contact **12**, and the driving rod **143** is fixed on the fixed seat **141** to push the fixed seat **141** to move. In this way, the driving effect of the driving member **14** is better, and when the control movable contact **12** is controlled to be disconnected from the stationary contact **11**, the response speed can be increased by the elastic member **142**.

(50) In one embodiment, the fixed seat **141** is configured as an insulator. Thus, the driving rod **143** is prevented from being electrified, and the working stability and use safety of the relay **100** are improved.

(51) In some embodiments, the relay body **10** further includes a fixed support **15**, the elastic member **142** is arranged in the fixed support **15**, a lower end of the fixed support **15** is connected to the fixed seat **141**, a guide groove **151** is formed in an upper end of the fixed support **15**, and a guide block **181** cooperating with the guide groove **151** is arranged on the first armature **18**.

(52) In this way, the movable contact **12** can stably contact with the stationary contact **11** under the pushing action of the elastic member **142** and the action of magnetic attraction force between the first armature **18** and the second armature, thus improving the working stability of the relay **10**. The structure of the relay **100** of the present application will be described below in one embodiment.

(53) The stationary contacts **11** are fixed on the housing **13** by brazing, the housing **13** is provided with the accommodating grooves **131**, the accommodating grooves **131** are used for fixing the magnetic members **22** and the covers **21**, the magnetic members **22** are arranged on two opposite sides of the cover **21**, a magnetic circuit formed by the two magnetic members **22** constitutes one arc extinguishing device **20**, and each stationary contact **11** is correspondingly provided with an arc extinguishing device **20**.

(54) In the housing **13**, the movable contact **12** is arranged at a position which is relatively isolated from the contact position of the stationary contact **11** by a certain distance (an electrical clearance), a stamping protrusion is arranged on the lower surface of the movable contact **12** and is used for

fixation with the first armature **18**, and at the same time, the riveting fixation of the movable contact **12** and the first armature **18** can also be achieved via the stamping protrusion.

(55) The first armature **18** is turned outwards on both sides to wrap the movable contact **12**, and the turn-up edge is pressed inwards to be pressed and fixed with the movable contact **12**. The fixed support **15** is a stamped and formed piece; the guide block **181** of the first armature **18** extends into the guide groove **151** of the fixed support **15**; the driving rod **143** is formed by machining or cold heading; the fixed support **15** is connected to the driving rod **143** via the fixed seat **141**; and the fixed seat **141** is configured as an insulator, does not have a direct contact with the fixed support **15**, and plays a role in high and low voltage insulation.

(56) A fixing position of the elastic member **142** is provided on the fixed support **15** and the fixed seat **141**; after the elastic member **142** is loaded, the first armature **18** riveted with the movable contact **12** is clamped into the fixed support **15**; two ends of the elastic member **142** respectively push against a positioning hole of the first armature **18** and a positioning structure of the fixed seat **141** so as to fix the first armature **18** and the movable contact **12** to the fixed support **15**; the guide block **181** extending out of the first armature **18** extends into a guide groove **151** reserved in the fixed support **15**; and when the relay **100** operates, the first armature **18** can move up and down along the guide groove **151**, thus, the connection and disconnection between the stationary contact **11** and the movable contact **12** of the relay **100**, and the response speed is increased.

(57) In the description of the present disclosure, it should be understood that, orientations or position relationships indicated by terms such as “center”, “longitudinal”, “transverse”, “length”, “width”, “thickness”, “up”, “down”, “front”, “back”, “left”, “right”, “vertical”, “horizontal”, “top”, “bottom”, “inner”, “outer”, “clockwise”, “counterclockwise”, “axial”, “radial”, and “circumferential” are orientations or position relationship shown based on the accompanying drawings, and are merely used for describing the present disclosure and simplifying the description, rather than indicating or implying that the apparatus or element should have a particular orientation or be constructed and operated in a particular orientation, and therefore, should not be construed as a limitation on the present disclosure.

(58) In the description of the present disclosure, a “first feature”, or a “second feature” can include one or more features.

(59) In the description of the present disclosure, “multiple” means two or more than two.

(60) In the description of the present disclosure, a first feature is “above” or “below” a second feature may include that the first feature directly contacts the second feature, or may include that the first feature does not contact the second feature directly but contacts the second feature through another feature between them.

(61) In description of the present disclosure, the first feature being “over”, “above”, and “on” the second feature includes that the first feature is directly above or obliquely above the second feature, or only means that a horizontal height of the first feature is higher than that of the second feature.

(62) In the descriptions of this specification, descriptions using reference terms “an embodiment”, “some embodiments”, “an exemplary embodiment”, “an example”, “a specific example”, or “some examples” mean that specific characteristics, structures, materials, or features described with reference to the embodiment or example are included in at least one embodiment or example of the present disclosure. In this specification, schematic descriptions of the foregoing terms do not necessarily point at a same embodiment or example. Moreover, the specific features, structures, materials, or characteristics described may be combined in any one or more embodiments or examples in an appropriate manner.

(63) Although the embodiments of the present disclosure have been shown and described, a person of ordinary skill in the art should understand that various changes, modifications, replacements and variations may be made to the embodiments without departing from the principles and spirit of the present disclosure, and the scope of the present disclosure is as defined by the appended claims and their equivalents.

Claims

1. A relay, comprising: a housing; a contact assembly which comprises movable contacts, stationary contacts and a movable contact bridge, wherein the movable contacts are adapted to move towards or away from the stationary contacts under the driving of the movable contact bridge; an auxiliary contact assembly which comprises an auxiliary contact bridge and auxiliary contacts, wherein at least portion of each of the auxiliary contacts extends into the housing, and the auxiliary contact bridge moves synchronously with the movable contact bridge to be selectively and electrically connected to the auxiliary contacts; wherein the housing comprises a top surface and stepped portions, the top surface is spaced apart from the stepped portion in the movement direction of the movable contact bridge or the auxiliary contact bridge, the stationary contacts are arranged on the top surface of the housing, and a portion of each of the auxiliary contacts arranged outside the housing is arranged on each of the stepped portions, the relay further comprises arc extinguishing devices, wherein the number of the arc extinguishing devices is multiple, each of the arc extinguishing devices is arranged on an outer side of the corresponding stationary contact and comprises a cover and a magnetic member, and the magnetic member is arranged on at least one of two side walls of the cover facing each other.
2. The relay of claim 1, wherein an area of the housing arranged between two adjacent stationary contacts is formed as a stepped portion, and the auxiliary contacts corresponding to the two adjacent stationary contacts are all arranged in the stepped portion.
3. The relay of claim 2, wherein the stepped portion is formed on a side wall of the housing, and the two auxiliary contacts corresponding to the two adjacent stationary contacts are arranged on the same stepped portion.
4. The relay of claim 3, wherein the number of the stationary contacts is multiple, the plurality of movable contacts electrically connected to each other are arranged on the movable contact bridge, and correspond to the plurality of stationary contacts on a one-to-one basis, and the movable contact bridge can selectively move towards the stationary contacts so as to conduct at least two of the plurality of stationary contacts; and the number of the auxiliary contacts is multiple, the plurality of the auxiliary contacts correspond to the plurality of stationary contacts on a one-to-one basis, and each of the auxiliary contacts extends out of the housing and is far away from the each of the stationary contacts.
5. The relay of claim 2, wherein the number of the stationary contacts is multiple, the plurality of movable contacts electrically connected to each other are arranged on the movable contact bridge, and correspond to the plurality of stationary contacts on a one-to-one basis, and the movable contact bridge can selectively move towards the stationary contacts so as to conduct at least two of the plurality of stationary contacts; and the number of the auxiliary contacts is multiple, the plurality of the auxiliary contacts correspond to the plurality of stationary contacts on a one-to-one basis, and each of the auxiliary contacts extends out of the housing and is far away from the each of the stationary contacts.
6. The relay of claim 1, wherein the number of the stationary contacts is multiple, the plurality of movable contacts electrically connected to each other are arranged on the movable contact bridge, and correspond to the plurality of stationary contacts on a one-to-one basis, and the movable contact bridge can selectively move towards the stationary contacts so as to conduct at least two of the plurality of stationary contacts; and the number of the auxiliary contacts is multiple, the plurality of the auxiliary contacts correspond to the plurality of stationary contacts on a one-to-one basis, and each of the auxiliary contacts extends out of the housing and is far away from the each of the stationary contacts.
7. The relay of claim 1, further comprising a driving member, wherein the driving member is connected to the movable contact bridge and is adapted to move the movable contacts towards or

away from the stationary contacts.

8. The relay of claim 7, wherein a first armature is arranged on the movable contact bridge, a second armature is arranged in the accommodating groove of the housing, and the first armature and the second armature are adapted to be magnetically attracted when the movable contact conducts the corresponding stationary contact.

9. The relay of claim 7, wherein the driving member comprises a fixed seat, an elastic member and a driving rod, one end of the elastic member is fixed on the fixed seat, the other end of the elastic member pushes against the movable contact, and the driving rod is fixed on the fixed seat to push the fixed seat to move.

10. The relay of claim 9, further comprising a fixed support, wherein the elastic member is arranged in the fixed support, a lower end of the fixed support is connected to the fixed seat, a guide groove is formed in an upper end of the fixed support, and a guide block cooperating with the guide groove is arranged on the first armature.

11. The relay of claim 9, wherein the fixed seat is configured as an insulator.
